

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec:Sr.Super60_NUCLEUS BT **Paper -2(Adv-2024-P2-Model)** **Date: 26-10-2025**

Time: 02.00Pm to 05.00Pm **GTA-04** **Max. Marks: 180**

26-10-2025_Sr.Super60_NUCLEUS-BT_Jee-Adv(2024-P2)_GTA-04_Syllabus,

MATHEMATICS: TOTAL SYLLABUS

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

Name of the Student: _____ **H.T. NO:** | | | | | | | |

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JEE-ADVANCED-2024-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 8 – 13)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 21)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 22 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 25 – 30)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 38)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 39 – 41)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 42 – 47)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

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MATHEMATICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

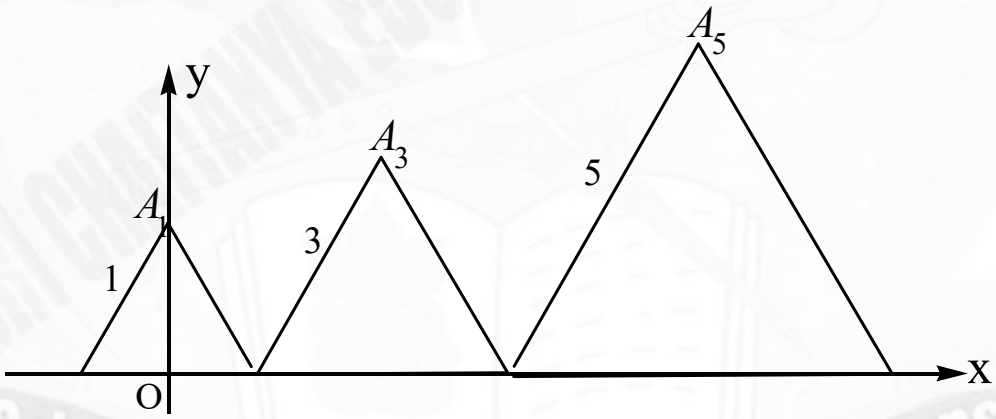
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

1. The value of $\left(\frac{{}^{20}C_0}{{}^{2}C_2}-\frac{{}^{20}C_1}{{}^{3}C_2}+\frac{{}^{20}C_2}{{}^{4}C_2}-\frac{{}^{20}C_3}{{}^{5}C_2}+....+\frac{{}^{20}C_{20}}{{}^{22}C_2}\right)$ is

A) $\frac{20}{11}$ **B)** $-\frac{1}{11}$ **C)** $\frac{1}{11}$ **D)** $\frac{23}{221}$

2. Equilateral triangle of side lengths 1, 3, 5, 7, …., $2n-1$ are placed end to end on x-axis with their base on the x-axis and mid-point of the base of the smallest triangle at origin. Vertices of the triangles not on x-axis are $A_1, A_3, A_5, A_7, \ldots\ldots, A_{2n-1}$ respectively.



The locus of the path formed by joining the vertices $A_1, A_3, A_5, \ldots\ldots, A_{2n-1}$ forms a conic whose latus rectum length is equal to

A) 2 **B)** 3 **C)** 4 **D)** 5

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3. If $f(x)=\frac{4x^2+1}{\sqrt{x^4+x^2}}+\frac{2x^4+2x^2+3}{\left(x^2+1\right)\sqrt{x^4+2}}$ and $y=f(x)$ for $x>0$ has least value for $x=\sqrt{\alpha}$ then the value of $\left(\sqrt{\frac{1}{\alpha}+1}-1\right)f\left(\sqrt{\alpha}\right)$ is

A) 1 **B)** 2 **C)** 0 **D)** 4

4. The unit's digit of $\prod_{k=0}^{97}\left(2+3\alpha_k^2\right)$, where $\alpha_0,\alpha_1,\alpha_2,.....,\alpha_{97}$ are the 98th root of unity is

A) 1 **B)** 0 **C)** 5 **D)** 6

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

5. Let A be a 3×3 invertible matrix with real entries such that $A^3-pA^2+p^2A=O$. Where p is a prime natural number then

A) No. of factors of $\left(-det\ (A)\right)$ are 4 **B)** Adjoint of A is square of A

C) Trace of AA^T is positive **D)** A^4 is a scalar multiple of A

6. Consider the equation as given $\frac{1}{x}+\frac{1}{y}=\frac{1}{p}$ where $x,y,p\in I^+$ and p is a prime number and (x,y) represents the solution pair then

A) Probability that $x<y$ is $\frac{1}{3}$ **B)** Probability that $x>y$ is $\frac{5}{6}$

C) Probability that $x\neq y$ is $\frac{2}{3}$ **D)** Probability that $x=y$ is $\frac{1}{6}$

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7. Let A, B are two points on the hyperbola $x^2+4xy-y^2-3=0$ with centre 'C' and the chord AB is bisected at (1,1). The line along the chord AB intersects its asymptotes in E and F. Then

A) Radius of the circumcircle of ΔCEF is $\sqrt{2}$

B) If e_k be the eccentricity of hyperbola $x^2+4xy-y^2-3k=0\forall k=1,2,3,.....2014.$

then $\prod_{k=1}^{2014} e_k = 2^{1007}$

C) Slope of the line perpendicular to the segment AE is $\frac{1}{3}$

D) Slope of the line perpendicular to the segment AE is -3.

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

8. If $\lim_{a\to\infty}\frac{1}{a}\int_0^{\infty}\frac{x^2+ax+1}{1+x^4}\tan^{-1}\left(\frac{1}{x}\right)dx=\frac{\pi^2}{k}, k\in N$, then $k=.....$

9. Considering the equation of the circle $x^2+y^2=9$ and a line $y=1$ which divides the circle Into two regions let small region be R_1 and other be R_2. Radius of biggest circle lying in region R_1 is P, Radius of biggest circle lying in region R_2 is Q. Then the value of P^2+Q^2 is

10. Angle between two focal chords of a parabola $(y-5)^2=8(x-1)$ which are tangents to the circle $x^2+y^2=9$ is $tan^{-1}\left(\frac{a}{b}\right)$, where a and b relatively prime numbers and Let A and B are two sets such that $A\cap B=\{a_1,a_2\}, A\cup B=\{a_1,a_2,a_3,a_4,a_5\}$ then maximum number possible functions from set A to set B is equal to c then $a-b+c$ is

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11. Given $f(z)=z^2-19z$, there are complex numbers z with the property that z, $f(z)$ and f(f(z)) are the vertices of a right triangle in the complex plane with a right angle at $f(z)$.there are positive integer m and n such that one such value of z is $m+\sqrt{n}+11i$,

$f(x)=|AB|$ Where $A_{3\times2}=\begin{bmatrix} x+1 & x+2 \\ 3x+3 & x+5 \\ 6x+2 & x+4 \end{bmatrix}; B_{2\times3}\begin{bmatrix} x+1 & x+2 & 5x+7 \\ x+2 & x+7 & 8x \end{bmatrix}$

Then the value of $(n-14m)+f(1)-2f'(1)+f''(1)=$_____

(where |M| denotes determinant of square matrix M)

12. Let there are 6 shirts of different colors and 6 trousers of same colors as that of shirts. If the number of ways in which these can be put on by 5 men such that no men wear the shirt and the trouser of the same color is $(k)6!$, then find the value of $(k-9)/5$ is

13. If $A=\sum_{r=1}^{3}\log_{(2+\sqrt{3})}\left|\tan\left(60^0-\alpha_r\right)\right|$, $B=\sum_{r=1}^{3}\log_{(2+\sqrt{3})}\left|\tan\left(60^0+\alpha_r\right)\right|$, where

$\alpha_r=\frac{\theta}{3^r}, r=1,2,3$ then the value of A + B at $\theta=\frac{9\pi}{4}$ is.....

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

Let $f(x)=x^3-12x^2+ax+b$ is an increasing function and $f(f(f(3)))=3$,

$f(f(f(f(4))))=4$ then

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14. The value of a is

15. The value of $f(7)$ is

PARAGRAPH-II

Let $\bar{a}=\bar{i}+2\bar{j}+3\bar{k}$; $\bar{b}=2\bar{i}+3\bar{j}+4\bar{k}$; $\bar{c}=4\bar{i}+\bar{j}$; $\bar{d}=5\bar{i}+2\bar{j}+\bar{k}$

Consider the sequence of vectors

$$\bar{x}_1=\bar{a}, \qquad \bar{x}_2=\bar{c}+\left(\frac{(\bar{x}_1-\bar{c}).\bar{d}}{|\bar{d}|^2}\right)\bar{d}, \qquad \bar{x}_3=\bar{a}+\left(\frac{(\bar{x}_2-\bar{a}).\bar{b}}{|\bar{b}|^2}\right)\bar{b},$$

$$\bar{x}_4=\bar{c}+\left(\frac{(\bar{x}_3-\bar{c}).\bar{d}}{|\bar{d}|^2}\right)\bar{d}, \qquad \bar{x}_5=\bar{a}+\left(\frac{(\bar{x}_4-\bar{a}).\bar{b}}{|\bar{b}|^2}\right)\bar{b}, \qquad \bar{x}_6=\bar{c}+\left(\frac{(\bar{x}_5-\bar{c}).\bar{d}}{|\bar{d}|^2}\right)\bar{d} \text{ and so on.}$$

16. If $\lim_{n\to\infty}\bar{x}_{2n}=\lambda_1\bar{i}+\lambda_2\bar{j}+\lambda_3\bar{k}$, $\lambda_1,\lambda_2,\lambda_3\in R$, then the value of $\frac{1}{2}\lambda_1-\frac{5}{3}\lambda_2+\frac{1}{3}\lambda_3=$

17. $\lim_{n\to\infty}\left|\bar{x}_{2n}-\bar{x}_{2n+1}\right|=$

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PHYSICS **Max Marks: 60**

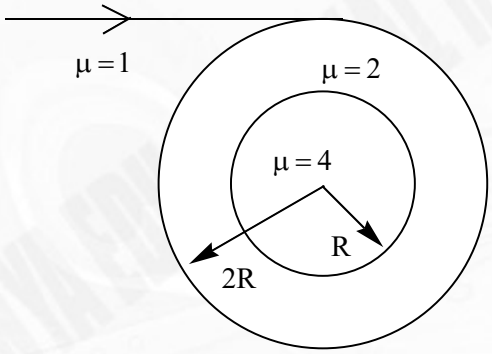
SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
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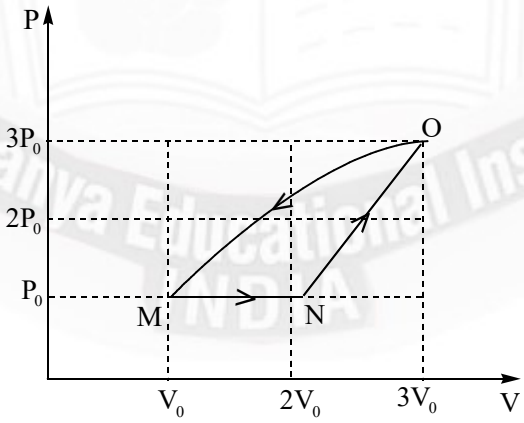
Full Marks : +3 If **ONLY** the correct option is chosen;
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Negative Marks : −1 In all other cases

18. Figure shows two spherical surfaces of radii R and $2R$ separating three transparent media of refractive index $\mu = 4, \mu = 2$ and $\mu = 1$. A ray of light travelling in medium $\mu = 1$ is incident on outer sphere tangentially. The net deviation suffered by light ray will be.



A) 0^0 **B)** 90^0 **C)** 240^0 **D)** 280^0

19. An ideal gas follows a cyclic process MNOM. In process MN heat absorbed by the gas is 200J where as in process OM internal energy of gas is decreased by 900J. The adiabatic constant for the gas will be



A) 2 / 3 **B)** 5 / 3 **C)** 16 / 9 **D)** 7 / 5

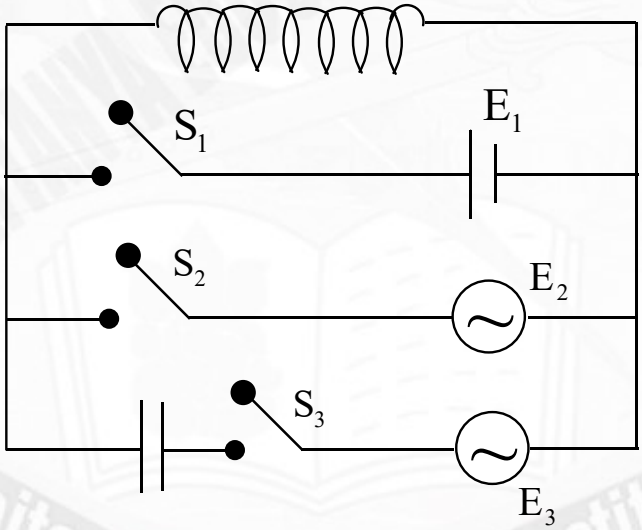
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20. In a setup of photoelectric experiment, a light of intensity $384\,W/m^2$ is used which is equally distributed among three wavelengths $620\,nm, 496\,nm$ and $310\,nm$. The light is incident perpendicular on a surface of area $0.01\,cm^2$, having work function of $2.25\,eV$. Assuming there is no loss of light and efficiency of emission of electrons from the surface is 50%. The photocurrent in the circuit is

A) 83.2×10^{-6} amp **B)** 166.4×10^{-6} amp

C) 20.8×10^{-6} amp **D)** 41.6×10^{-6} amp

21. In the given circuit $E_1 = 24$ volt DC, $E_2 = 24$ volt AC, 50rad/s, and $E_3 = 12$ volt AC, 50 rad/s and $C = \frac{1}{50}F$. With S_1 is closed and S_2, S_3 are open, a steady current 8 amp is observed. With S_1, S_3 open and S_2 closed current observed is 4.8 amp . Mark the correct answer.



A) Inductance of coil is nearly is 0.8 H

B) Resistance of coil is 6Ω

C) If S_3 closed with S_1 and S_2 open rms value of current in the circuit will be 2 A

D) If S_3 is closed with S_1 and S_2 are open, average power supplied by battery will be 24W

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
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22. A small pendulum is suspended from a peg on a vertical wall. The pendulum is pulled away from the wall to a horizontal position and released. The ball hits the wall, the coefficient of restitution being $2/\sqrt{5}$. If mass of the bob is m then choose the correct option

A) Maximum angle with vertical will the pendulum make after the first collision with the wall is $\cos^{-1}\left(\frac{1}{5}\right)$

B) Maximum angle with vertical will the pendulum make after the first collision with the wall $\cos^{-1}\left(1-2/\sqrt{5}\right)$

C) After 3 number of collisions bob will raise height of $\frac{64L}{125}$

D) After 4 number of collision bob will raise up to height of $64L/25$

23. A wave pulse starts propagating in positive x direction along a non uniform wire of length ℓ, with mass per unit length given by $\mu=(\mu_0+\alpha x)\frac{kg}{m}$. Wave pulse starts from $x=0$ to heavier end. Tension in wire is 100 N.

A) Wave velocity as a function of x is given by $v=\frac{10}{(\mu_0+\alpha x)^{\frac{1}{2}}}$

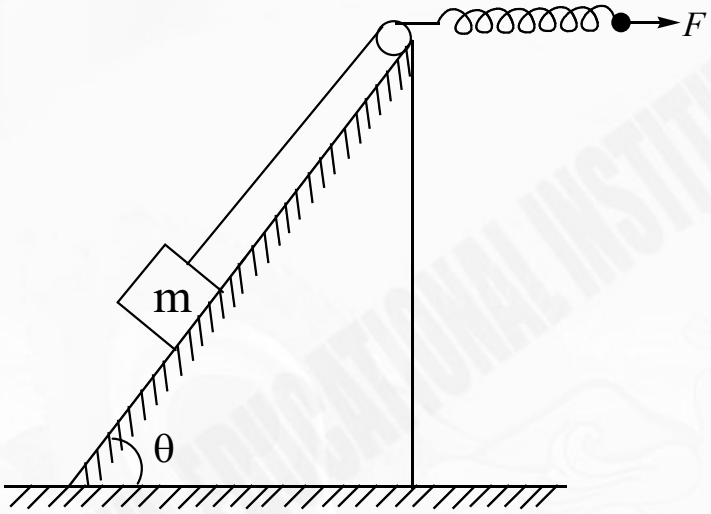
B) Wave speed decreases by increasing x coordinate

C) Wave frequency remains same as wave propagates

D) Time taken by wave pulse to cross the string is $t=\frac{1}{15\alpha}\left[(\mu_0+\alpha l)^{3/2}-\mu_0^{3/2}\right]$

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24. A block of mass m rests on a rough incline with coefficient of friction μ as shown. It is known that $\mu > \tan\theta$. The block is attached with a spring of spring constant k through a light rope that passes over a pulley. The other end of the spring is pulled with a gradually increasing force F.



A) Elongation in the spring is $\frac{mg\sin\theta}{k}$ when friction force between block and inclined plane is zero.

B) Elongation in the spring is $\frac{mg\sin\theta + F}{k}$ when friction force between block and inclined plane is zero.

C) Potential energy stored in the spring when block m begins to move is $\frac{m^2g^2(\sin\theta + \mu\cos\theta)^2}{2k}$

D) Work done by force F till time when block m starts to move is $\frac{m^2g^2(\sin\theta + \mu\cos\theta)^2}{k}$

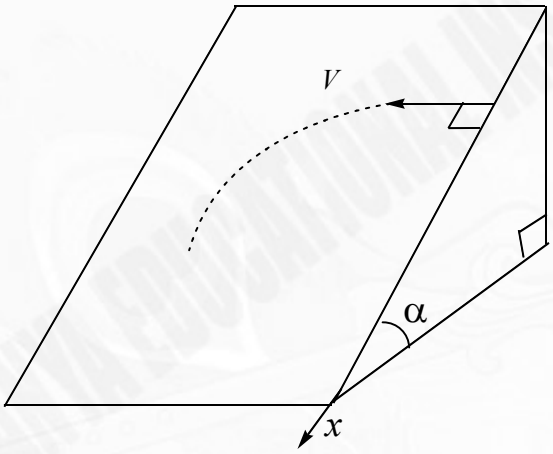
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SECTION-III
(NON-NEGATIVE INTEGER)

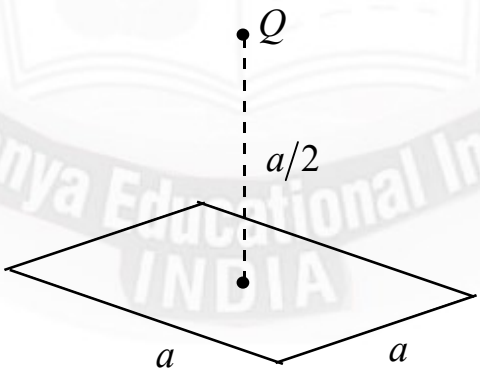
- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases..

25. There is a sufficiently large fixed inclined plane making an angle α with the horizontal. A small block is given a velocity of $64\ \text{m/s}$ perpendicular to the $x-axis$ (which is parallel and directed down the incline) in the plane of the incline, the friction coefficient between block and incline surface is given by $\mu = \tan\alpha$ where α is the inclination of the plane. Find the magnitude of velocity of block after its long-time motion. (in m/s)

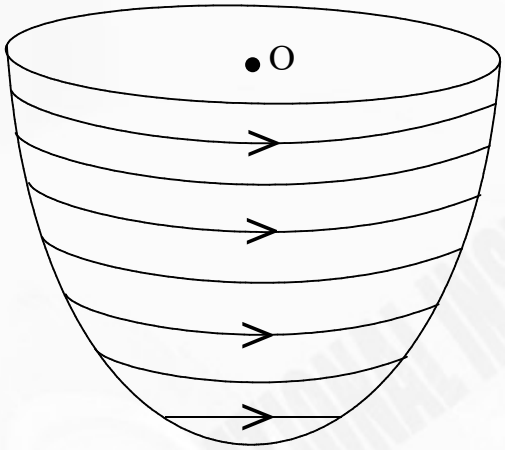


26. Square plate of side length a is placed at a normal distance $\frac{a}{2}$ from a point charge Q is placed symmetrically as shown. If uniform charge density of square plate is σ, then square plate experience the force of $\frac{\sigma Q}{n\varepsilon_0}$.The value of n is



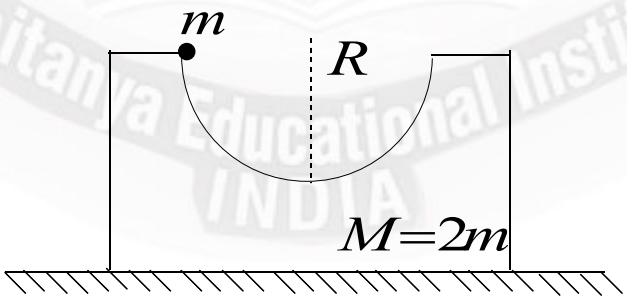
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27. A wire is wrapped tightly without any gap over a hemispherical bowl of radius R .If N is the total number of turns and i is the current per turn, the magnetic field at the centre O will be $\frac{\mu_0 Ni}{\alpha R}$ Find the value of α .



28. A radioactive sample can decay by either of two processes X and Y. The half life for process X is $\frac{1}{4}hr$ and for process Y is $\frac{1}{6}hr$. In a hypothetical process the sample decays for first half hour by process X, for next one hour by process Y and for next half hour by processes X and Y both. The initial number of nuclei of the sample is N_0. After 2 Hr, the number of undecayed nuclei of the sample are $\frac{N_0}{32}\left(\frac{1}{2}\right)^n$. Find the value of n.

29. Wedge of mass $M = 2m$ has hemispherical groove of radius as $R = 9cm$ as shown in figure. A particle of mass m kept at upper extreme of groove released from rest. Find the radius of curvature in cm of the particle, when it reaches its lowermost point. (All surfaces are smooth) (Take $g = 10m/s^2$)



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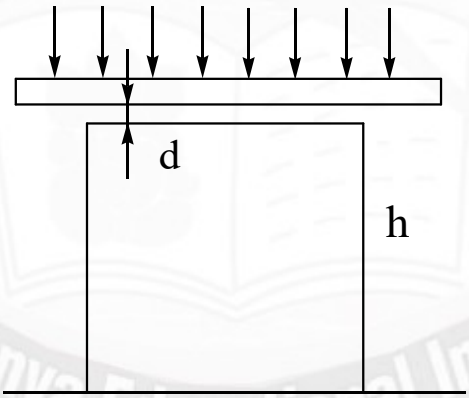
30. Two infinite long wires are having uniform linear charge density $+\lambda$ and $-\lambda$, respectively. They are parallel to $z-$axis and passes through $x-$axis at $x_1=-a$ and $x_2=+a$ respectively. The radius of an equipotential cylinder having potential $\frac{\lambda \ln(2)}{4\pi\varepsilon_0}$ volt is found $ka\sqrt{2}$. Find the value of k.

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

A glass plate is placed above a glass cube of 2 cm edges in such a way that there remains a thin air layer between them, see figure. Electromagnetic radiation of wavelength between 400 nm and 1150 nm (for which the plate is penetrable) incident perpendicular to the plate from above is reflected from both air surfaces and interferes. In this range only two wavelengths give maximum reinforcements, one of them is 400 nm . λ is the second wavelength. T is change in temp. necessary to warm



up the cube so as it would touch the plate. The coefficient of linear thermal expansion is $\alpha=8\times10^{-6}/^{0}\text{C}$. the refractive index of the air $\mu=1$. The distance of the bottom of the cube from the plate does not change during warming up.

Then,

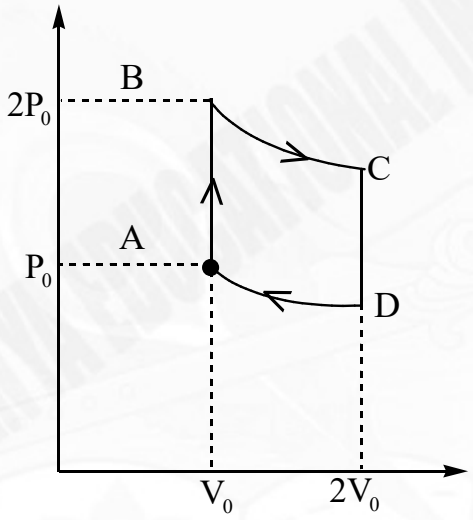
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31. The value of λ (in nm)

32. The value of $T\left(\text{in}^{0}\text{C}\right)$

PARAGRAPH-II

One mole of a monatomic gas undergoes the cycle shown in the diagram: The pressure, volume and temperature at A are P_0, V_0 and T_0 respectively; the pressure at B is $2P_0$ and the volume at D is $2V_0$. The processes AB and CD are isochoric. During BC DA, the gas undergoes the processes such that during an infinitesimal contraction or expansion, the heat absorbed dQ, the change in internal energy dU and dW the work done satisfy the relation:



$dQ = \frac{1}{2}dU + \frac{2}{3}dW$.

For the process $BC\ DA$ we can write

$dQ = \frac{1}{2}dU + \frac{2}{3}dW = dU + dW$.

Or, $+\frac{1}{2}dU = -\frac{1}{3}dW$; Or, $\frac{1}{2}\cdot\frac{3}{2}RdT = -\frac{1}{3}dW$

$pdV + Vdp \Rightarrow RdT$; Or, $\frac{13}{9}pdV + Vdp = 0$

Or, $pV^{\frac{13}{9}} = \text{constants}$

33. If temperature at C is xT_0. Find the value of ' X '.

34. What is the efficiency of the cycle? Assume $2^{5/9} = 1.5, 2^{4/9} = 4/3$ approximately.

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CHEMISTRY **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

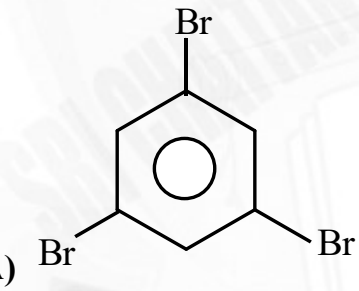
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

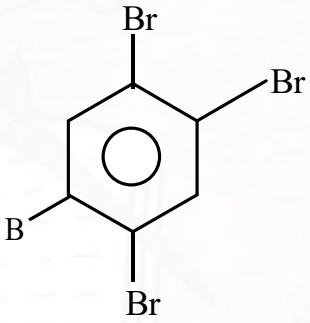
35. Toluene reacts with mixed acid $(concHNO_3\ and\ conc\ H_2SO_4)$ at $288K$ to give $P(59\%)$, $Q(37\%)$, $R(2\%)$. The major product(s) of the following reaction sequence is (are)

$$P \xrightarrow[(3)\ KMnO_4, \bar{O}H, \Delta \\ (5)\ H_3O^{\oplus}]{(1)\ Br_2, CH_3COOH} S \xrightarrow[(2)\ Sn/HCl \\ (3)\ Br_2/H_2O \\ excess \\ (4)\ NaNO_2, HCl \\ 273-278k \\ (5)\ H_3PO_2]{(1)\ NaOH+CaO, \Delta} major\ product(T)$$

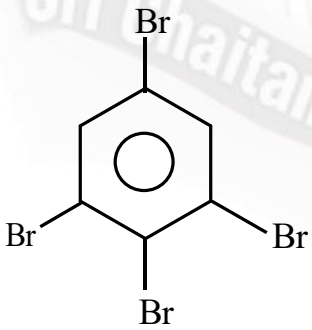
A)



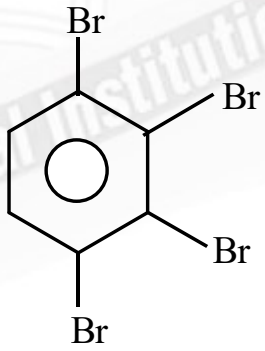
B)



C)

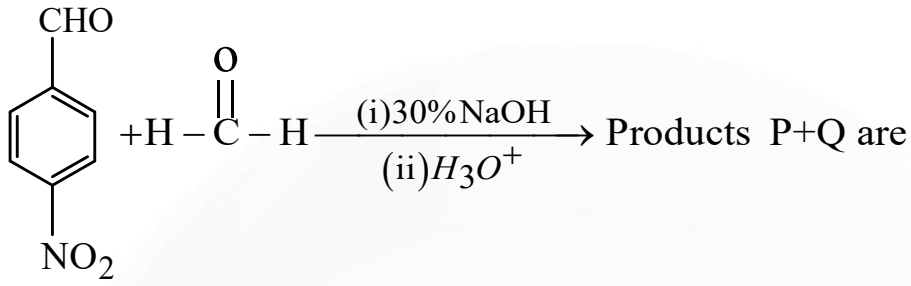


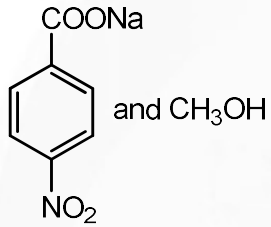
D)

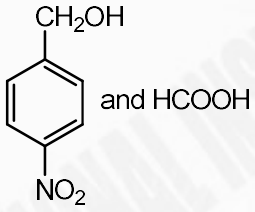


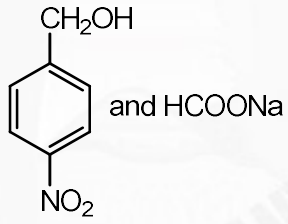
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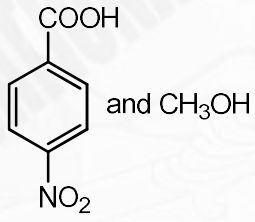
36.



A) 

B) 

C) 

D) 

37. The conductivity of 10^{-3}M CH_3COOH solution at 'T' K is 5×10^{-5} Scm^{-1}. The Osmotic pressure of the Aqueous solution at 'T' K is:

$\left(\Lambda_m^\circ CH_3COONa = 96\ Scm^2\ mol^{-1}\ ;\ \Lambda_m^\circ HCl = 430\ Scm^2\ mol^{-1}\ ;\ \Lambda_m^\circ NaCl = 126\ Scm^2\ mol^{-1}\ \text{at 'T' K}\right)$

A) $1.125\times RT$ **B)** $2.25\times10^{-3}RT$ **C)** $2.25\times RT$ **D)** $1.125\times10^{-3}RT$

38. Incorrect statement among the following is

A) $KOH_{(aq)}$ is better absorber of CO_2 gas than $NaOH(aq)$.

B) $E^0_{M^+/M}$ value for lithium is more negative than sodium hence lithium is more reactive with water than sodium.

C) When $NaNO_3$ is heated to 800^0C; Na_2O, N_2 and O_2 are formed.

D) When $NaNO_3$ is heated to 500^0C; $NaNO_2$ and O_2 are formed.

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

39. Which of the following orders is/are correct

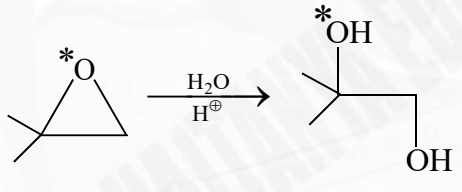
A) $HOCl > HClO_2 > HClO_3 > HClO_4$ (oxidizing ability)

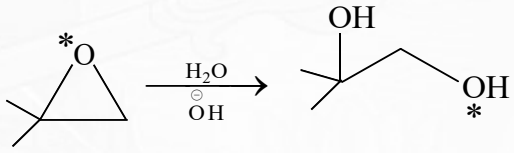
B) $XeF_2 > XeF_4 > XeF_6$ (melting point)

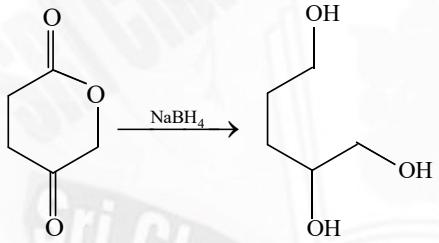
C) $SF_2 > SF_4 > SF_6$ [formal charge on S-atom]

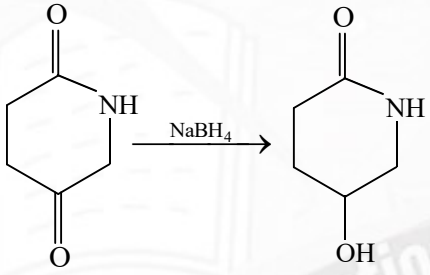
D) $CO > NO$ (Ist Ionization energy)

40. Select incorrect statements

A) 

B) 

C) 

D) 

41. 20 ml of a H_2O_2 solution requires 50 ml of $0.2N$ $KMnO_4$ for complete titration in acidic medium then for H_2O_2 solution:

A) Normality = 0.5N

B) Molarity = 0.25M

C) Volume strength of $H_2O_2 = 2.8v$ (at 1 atm pressure)

D) Volume of O_2 (in l) evolved at NTP by 2 litre of H_2O_2 solution is 5.6 *liters*

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SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

42. Among the species given below, The total number of species which is repelled by external magnetic field

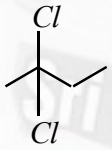
Na in vapour phase (at low concentration and high temperature), $NO(g)$, O_2^{-2} (peroxide), Diatomic sulphur in vapour phase, $\left[Co(H_2O)_3F_3\right]$, K_2MnO_4, $(NH_4)_2[NiCl_4]$, $K_2[PtCl_4]$

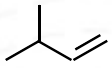
43. A certain mass of a substance when dissolved in 100 g C_6H_6 lowers the freezing point by 1.28^oC. The same mass of solute dissolved in 100 g of water lowers the freezing point by 1.40^oC. If the substance has normal molecular weight in benzene and is completely dissociated in water, into how many ions does it dissociate in water? K_f for H_2O and C_6H_6 are 1.86 and 5.12 $K\ mol^{-1}\ kg$ respectively.

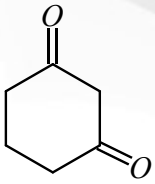
44. A galvanic electrochemical cell made of Zn^{+2}/Zn *and* Cu^{2+}/Cu half-cells produces 1.10V at $25°C$. When the ratio of $\left[Zn^{2+}\right]$ *to* $\left[Cu^{2+}\right]$ is maintained at 1.0. The $\Delta G°$ for the reaction when 1.0 *mole of* Zn get dissolved is $-53x\ kJ$. Then the value of 'x' is: (round off to the nearest integer)

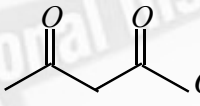
[Given: Faraday's constant $= 96485\ C\ mol^{-1}$]

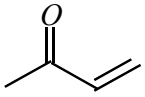
45. How many of the following reactions give haloform as one of the products?

(i)  $\xrightarrow[(ii)\ I_2/NaOH]{(i)\ H_2O/\Delta}$

(ii)  $\xrightarrow[(ii)\ I_2/NaOH]{(i)\ H_3O^+}$

(iii)  $\xrightarrow{I_2/NaOH}$

(iv)  $\xrightarrow{I_2/NaOH}$

(v)  $\xrightarrow[(ii)\ I_2/NaOH,\Delta]{(i)\ Hg(OAC)_2/H_2O,\ NaBH_4}$

(vi) $Ph-CH_2-\overset{\overset{O}{\|}}{C}-CH_3 \xrightarrow{I_2/NaOH}$

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46. For the cyclic silicate ion with three silicon atoms, the number of oxygen atoms shared is x and the magnitude of overall charge present is y. Then the value of y-x is:

47. The rate of a reaction starting with initial concentration of reactant is $2\times10^{-3}M$ and $1\times10^{-3}M$ are equal to $2.40\times10^{-4}M\ sec^{-1}$ and $0.60\times10^{-4}M\ sec^{-1}$ respectively. Calculate the order of reaction w.r.t reactant. [log 2.4 = 0.38 log 2 = 0.30, log 3 = 0.48]

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

Two flasks of equal volume, connected by a narrow tube of negligible volume contain 1.0 mole of H_2 gas at 300 K and 0.5 atm. Now one of the flasks is immersed into a thermostat maintained at 450 K and other was maintained at constant 300 K temperature

48. Amount of H_2 gas in flask maintained at 300 K finally is ____gm

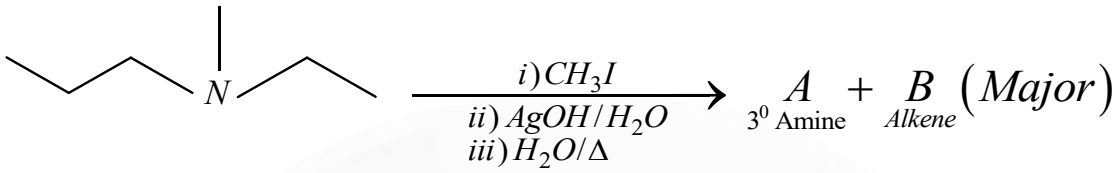
49. Final pressure in both the vessels is _____ atm

PARAGRAPH-II

Tertiary amine on reaction with methyl iodide gives quaternary ammonium iodide which on reaction with Ag_2O/H_2O gives 4^0 ammonium hydroxide and on heating finally gives 3^0 amine and alkene. This reaction is called Hoffmann elimination. (if alkyl groups attached to N' contains β hydrogens).

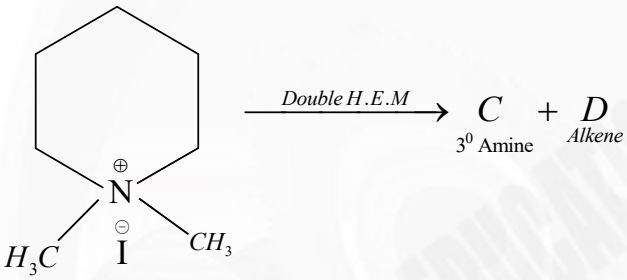
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50.



If 'B' contains 'P' number of carbon atoms and 'Q' number of hydrogen atoms then $\frac{P}{Q}$ is _____

51.



(H.E.M=Hofmann exhaustive methylation)

If the molar mass of alkene is 'R'. Then $\frac{R}{10}$ is _____

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